EHB TECHNOLOGIES · HUMAN RESOURCES

HR Self-Service Portal

Project overview: what we are building and how it works

13 October 2026 · Version 1 · Rules v1 awaiting management sign-off

In one line

A web portal where every EHB employee handles their own HR work — attendance, leave, pay slips, requests and documents — and every request moves automatically to the right approver, with HR, Finance and management seeing the same up-to-date records.

Users	All EHB employees, plus Line Managers, HODs, HR, Finance, Admin and Management
Platform	Web application (works on phone browsers); no separate mobile app
Scope	Dashboard, attendance, leave, pay slips, 13 request forms, documents, performance, assets
Rollout	Three phases; Phase 1 covers the daily essentials
Estimated build	About 10–14 weeks for all phases; Phase 1 in about 3 weeks (estimate)
Rules	Defined in the companion document “HR Portal — Final Rules v1”

Why we are building it

- Today, leave, attendance corrections, letters and claims move by email, paper or messages, so requests get lost and nobody sees their status.
- Employee records are spread across Google Forms and spreadsheets.
- Managers and HR have no single view of who is on leave, who is late, or which requests are waiting.
- Employees cannot see their own leave balance, pay slips or PF statement without asking HR.

1. Who uses the portal

Every user logs in with their own account. What they see depends on their role; one person can hold more than one role (for example, a Line Manager is also an employee).

Role	What they do in the portal
Employee	Sees own dashboard, attendance, leave balance and pay slips; applies for leave and other requests; updates own contact details; downloads policies
Line Manager	Everything an employee does, plus approves or returns team requests and sees the team's attendance and leave
Head of Department (HOD)	Second-level approval for expenses, travel, resignations and hiring requests
HR	Manages employee records, holiday calendar, leave quotas and policies; imports attendance; final approval where the rules say HR
Finance	Uploads monthly payroll; approves training, expense, travel and PF payouts
Admin	Books travel, handles visiting cards and company assets
Management / CEO	Views reports and salary data; sits on the PF Board / Committee

2. What the portal includes

Phase	Modules	What the user gets
1 — at launch	Login and dashboard · Employee profiles · Attendance · Attendance Update · Leave · Pay Slip and tax statement · Policies and HR documents	The daily essentials: check in-time and leave balance, apply for leave, fix a missed punch, download a pay slip
2 — next	Letter Request · Expense Claim · Travel · Resignation and exit clearance · Performance (BSC) · My Inventory (assets) · Manpower Requisition and internal jobs	All other common requests and the six-monthly performance cycle
3 — later	PF statement and withdrawal · Training request · Visiting card	Less frequent requests
Not included	Loan · Advance Salary · Award Reimbursement	EHB does not offer these today

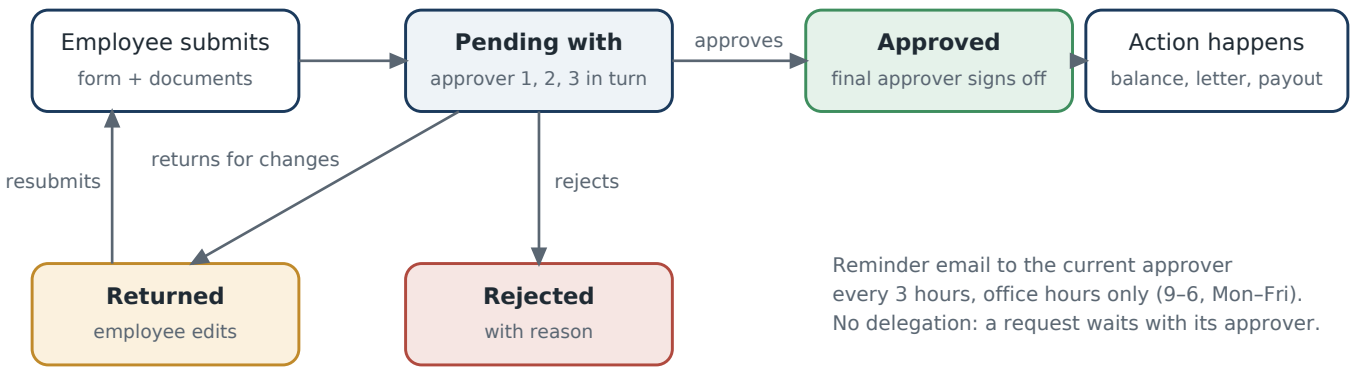
3. How it works

3.1 A normal day for an employee

1. **Log in** with work email and password.
2. **Dashboard** shows today's in-time, this month's attendance (On Time, Late, Half Day, Absent), leave balances, pending requests and any tasks waiting for their approval.
3. **Apply** for something (leave, letter, expense) from the Requests menu. The form already knows their name, department and manager.
4. **Track** the request: status changes from Pending to Approved, Rejected or Returned, and they get an email at each step.
5. **Download** pay slips, tax statement, policies and letters whenever needed.

3.2 How every request is approved

All 13 request forms use the same approval engine. Each request type has its own chain of approvers (for example, an expense claim goes Line Manager → HOD → Finance), but the steps are always the same:



Every step is recorded: who approved, when, and any comment. The employee always sees where the request is.

3.3 Attendance

- The biometric machine records check-in and check-out. HR exports the data (Excel / CSV) and imports it; the portal matches each machine ID to an employee code.
- The portal works out each day's status from the rules: arrival after 9:25 AM is Late; 6+ hours Present; 4 to under 6 hours Half Day; under 4 hours or no record without leave is Absent. Weekends and public holidays are skipped.
- Three unapproved late arrivals in a month raise a warning to HR; there is no automatic deduction.
- A missed punch or field visit is fixed with an Attendance Update request (up to 3 a month, within 7 days). The original record is never overwritten.

3.4 Leave

- The employee picks dates and leave type. The portal counts working days (skipping weekends and holidays), checks the balance and the notice period, and asks for a document when needed (e.g. a medical certificate for more than 2 sick days).
- The request goes to the Line Manager (and HR for leave without pay). On approval the balance updates and the days show on the attendance calendar.

- Quotas follow the calendar year and are pro-rated for new joiners; up to 14 unused Annual Leave days carry forward.

3.5 Pay slips and tax statement

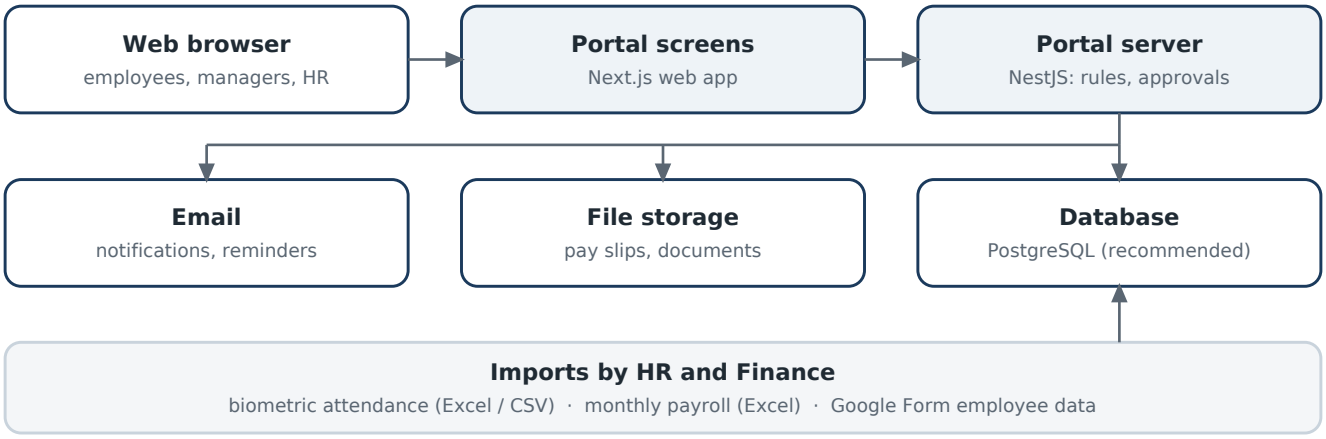
- The portal does not calculate salary. Each month Finance uploads the payroll Excel file.
- The portal creates a pay slip for each employee showing earnings (Basic, House Rent, Medical, Conveyance, Fuel, Mobile) and deductions (Income Tax, EOBI, Provident Fund), available by the 5th.
- Employees download pay slips as PDF and view a yearly tax statement for July–June.

3.6 Employee records and documents

- Starting records are imported from the Employee Personal Information Google Form; each employee gets a code such as EHB-2026-001.
- Employees update their own contact details; HR changes job, department, manager and salary fields.
- HR uploads policies and HR forms as PDFs; employees read and download them from the portal.

4. How the system is built

The portal has three layers: the screens people use, a server that applies the rules, and a database that keeps the records. Spreadsheets from the biometric machine, payroll and the Google Form come in through HR and Finance uploads.



Part	Technology	Why
Screens	Next.js + React + TypeScript, Tailwind CSS	The stack our team already uses; fast, works on phone browsers
Server	NestJS + TypeScript	Clear modules per feature; same language as the screens
Database	PostgreSQL (recommended)	HR data is highly related (employee → manager → request → approvals) and needs reliable transactions; final choice pending
Files and PDFs	Cloud file storage; PDF generation	Pay slips, letters and uploaded documents
Email and reminders	Email service + background job queue	Step notifications and the 3-hour reminders
Hosting	Docker on a cloud server, HTTPS	Simple to deploy and back up

5. Security and privacy

- Each person sees only what their role allows. Line Managers see work information of their own team only.
- CNIC and family details are visible to authorised HR staff only; bank details to HR / payroll only; medical documents to HR only.
- Salary data is visible to the employee, HR, Finance and the CEO.
- All changes to records and all views of sensitive data are logged with who and when.
- Passwords are stored encrypted; all traffic uses HTTPS; data is backed up daily.

6. Timeline and next steps

Step	What happens	Estimated time
Rules sign-off	HR Manager, Finance and Management approve “Final Rules v1” and settle its 10 open items	1–2 days, in parallel with the build
Phase 1 — foundation	Project setup, employee records import, login and roles, approval engine, first form (Letter Request)	About 1.5 weeks
Phase 1 — modules	Dashboard, attendance import, Attendance Update, Leave, Pay Slip, Policies; testing with HR	About 1.5 weeks
Phase 2	Remaining requests, exit clearance, performance (BSC), assets, hiring	About 4–6 weeks
Phase 3	PF, training, visiting cards; admin settings polish	About 2–3 weeks

Estimates assume one developer working about 6 hours a day with AI-assisted development; they will be refined after Phase 1. Every quota, limit and day count is a setting, so HR can change rules later without new development.

Decisions needed now

- Sign-off on Final Rules v1, including maternity and paternity days (to be checked against the federal Maternity and Paternity Leave Act).
- Biometric machine brand, model and export format (IT / Admin).
- Database choice: PostgreSQL (recommended) or MongoDB.